

Figure Matrix

Matrix is a combination of Row and column. In this type of questions two matrices of letters and numbers are given in which each letters can be represented by a set of two numbers. The first number (from left) indicates the row number while the second number indicates the column number. You are required to identify the code for given word or a group of letters on the basis of given two matrices. If any two digits written as '45' it means 4 is number of row and 5 is number of column.

Solved Examples

Directions (1-2): A word is represented by only one set of numbers as given in anyone of the alternatives. The sets of numbers given in the alternatives are represented by two classes of alphabets as in two matrices given below. The columns and rows of Matrix I are numbered from 0 to 4 and that of Matrix II are numbered from 5 to 9. A letter from these matrices can be represented first by its row and next by its column. e.g. A can be represented by 01, 20, 42 etc. and H can be represented by 65, 57, 98 etc.

Similarly, you have to identify the set for the word given in the question.

MATRIX-I

	0	1	2	3	4
0	F	A	N	C	I
1	I	O	F	A	N
2	A	N	O	I	F
3	O	F	I	N	A
4	N	I	A	F	O

MATRIX-II

	5	6	7	8	9
5	S	E	H	B	T
6	H	S	E	T	B
7	B	T	S	E	H
8	E	H	T	B	S
9	T	S	E	H	B

1. (FAITH)

- (a) 24, 31, 10, 59, 57
- (b) 12, 20, 40, 68, 65
- (c) 31, 34, 23, 76, 79
- (d) 43, 42, 41, 78, 89

Sol. F $\Rightarrow$ 00, 12, 24, (31), 43

A $\Rightarrow$ 01, 13, 20, (34), 42

I $\Rightarrow$ 04, 10, (23), 32, 41

T $\Rightarrow$ 59, 68, (76), 87, 95

H $\Rightarrow$ 57, 65, (79), 86, 98

FAITH $\Rightarrow$ 31, 34, 23, 76, 79

2. TOFAN

- (a) 68, 22, 00, 02, 02
- (b) 59, 11, 31, 13, 14
- (c) 76, 44, 43, 33, 67
- (d) 87, 30, 12, 42, 98

Sol. T O F A N 59 11 31 13 14

3. Read and study the following matrix and explain how the words will be coded ?

	1	2	3	4	5
1	H	N	A	S	D
2	D	R	M	S	A
3	R	M	S	D	H
4	N	A	D	M	H
5	S	A	N	D	H

HANDS

- (a) 35, 42, 53, 21, 14
- (b) 35, 42, 53, 21, 23
- (c) 35, 42, 53, 21, 43
- (d) 11, 35, 42, 53, 21

Sol. We find the code of the word 'hands'

H = 11, (35), 45, 55 A = 13, 25, (42), 52

N = 12, 41, (53) D = 15, (21), 34, 43, 54

S = (14), 24, 33, 52

Thus the correct code of the word 'HANDS' is 35, 42, 53, 21, 14

4. Which of these columns are co-related and how?

Line	I	II	III	IV	V
1	20	28.2	10	50	30
2	25	30	12.5	60	37.5
3	80	41	40	90	120
4	26	70	13	45	39
5	30	80	15	80	45

- (a) Column I, II, V
- (b) Column II, III, IV
- (c) Column I, III, IV
- (d) Column I, III, V

Sol. (d); Column I, III, V

I	II	V
$20 \div 2$	10×3	30
$25 \div 2$	12.5×3	37.5
$80 \div 2$	40×3	120
$26 \div 2$	13×3	39
$30 \div 2$	15×3	45

5. Among the four sets, find out the set, which is like given set.

Given set (12, 72, 216)

- (a) (4, 24, 48) (b) (7, 42, 252)
(c) (11, 60, 30) (d) (5, 30, 90)

Sol. (d); The rule in given set is

$$12 \times 6 = 72$$

and $72 \times 3 = 216$

Similarly, $5 \times 6 = 30$

and $30 \times 3 = 90$

Practice Set

Directions (1-2): In the following questions given below are two matrices, each containing two classes of letters from the alphabet. The columns and rows of matrix I are numbered from 0 to 4 and that of matrix II from 5 to 9. A letter from these matrices can be represented first by its row number and next by its column number. For example, R can be represented by 02, 31. In each of the following questions identify one set of number pairs out of (a), (b), (c), (d) which represents the given word:

MATRIX-I

	0	1	2	3	4
0	E	S	R	U	N
1	R	N	S	E	U
2	U	E	N	R	S
3	S	R	U	N	E
4	N	U	E	S	R

MATRIX-II

	5	6	7	8	9
5	W	O	P	T	I
6	T	I	O	W	P
7	O	W	I	P	T
8	I	P	T	O	W
9	P	T	W	I	O

1. PENT
(a) 87, 21, 31, 66 (b) 95, 33, 40, 78
(c) 57, 02, 34, 87 (d) 78, 42, 11, 58
2. NOTE
(a) 40, 75, 96, 34 (b) 33, 99, 87, 14
(c) 04, 67, 78, 21 (d) 22, 56, 65, 43

Directions (3-7):

MATRIX-I

	0	1	2	3	4
0	F	O	M	S	R
1	S	R	F	O	M
2	O	M	S	R	F
3	R	F	O	M	S
4	M	S	R	F	O

MATRIX-II

	5	6	7	8	9
5	A	T	D	I	P
6	I	P	A	T	D
7	T	D	I	P	A
8	R	A	T	D	I
9	D	I	P	A	T

3. ROAD
(a) 04, 20, 55, 78 (b) 23, 32, 98, 99
(c) 42, 32, 79, 58 (d) 11, 13, 67, 69
4. MOST
(a) 02, 13, 34, 56 (b) 21, 00, 03, 88
(c) 33, 20, 11, 79 (d) 40, 44, 22, 89

5. FOAM

(a) 24, 01, 55, 22

(c) 12, 13, 67, 23

6. STOP

(a) 10, 56, 44, 97

(c) 22, 75, 32, 86

7. MOST

(a) 02, 31, 34, 75

(c) 41, 01, 42, 68

8. FORK

MATRIX-I

	0	1	2	3	4
0	M	R	H	N	F
1	N	F	M	R	H
2	R	H	N	F	M
3	F	M	R	H	N
4	H	N	F	M	R

(a) 11, 79, 20, 67

(c) 20, 96, 32, 55

MATRIX-II

	5	6	7	8	9
5	O	E	A	K	U
6	K	U	O	E	A
7	E	A	K	U	O
8	U	O	E	A	K
9	A	K	U	O	E

(b) 30, 86, 13, 77

(d) 23, 86, 11, 77

Directions (9-13):

MATRIX-I

	0	1	2	3	4
0	A	B	C	D	E
1	B	C	D	E	A
2	C	D	E	A	B
3	D	E	A	B	C
4	E	A	B	C	D

MATRIX-II

	5	6	7	8	9
5	P	Q	R	S	T
6	Q	R	S	T	P
7	R	S	T	P	Q
8	S	T	P	Q	R
9	T	P	Q	R	S

9. REST

(a) 66, 13, 95, 33

(c) 89, 22, 76, 86

10. QATAR

(a) 22, 04, 59, 67, 75

(c) 79, 95, 98, 75, 99

11. RADAR

(a) 75, 32, 03, 41, 57

(c) 89, 32, 30, 42, 75

12. READ

(a) 75, 13, 32, 65

(c) 79, 18, 42, 56

(b) 55, 66, 21, 42

(d) 89, 22, 36, 69

(b) 65, 00, 33, 14, 57

(d) 97, 41, 59, 14, 75

(b) 76, 24, 30, 15, 78

(d) 89, 00, 30, 42, 98

(b) 75, 13, 32, 12

(d) 75, 03, 32, 43

13. TEAR

- (a) 95, 40, 32, 72
(c) 95, 40, 23, 66

- (b) 95, 58, 90, 66
(d) 66, 58, 95, 23

Direction (14-16):

MATRIX-I

	0	1	2	3	4
0	N	K	R	C	B
1	R	N	C	B	K
2	K	R	B	N	C
3	B	C	N	R	K
4	C	B	K	N	R

MATRIX-II

	5	6	7	8	9
5	O	A	I	U	E
6	I	E	O	A	U
7	A	O	E	I	U
8	U	O	A	E	I
9	E	I	U	O	A

14. RICE

- (a) 02, 65, 31, 41
(c) 77, 65, 34, 10

- (b) 10, 96, 24, 77
(d) 88, 97, 40, 03

15. NIKE

- (a) 34, 75, 01, 95
(c) 34, 57, 24, 66

- (b) 23, 69, 24, 55
(d) 11, 78, 34, 77

16. ROAR

- (a) 50, 67, 86, 02
(c) 21, 67, 65, 80

- (b) 44, 05, 98, 44
(d) 33, 55, 87, 21

Direction (17-18):

MATRIX-I

	0	1	2	3	4
0	E	S	R	U	N
1	R	N	S	E	U
2	U	E	N	R	S
3	S	R	U	N	E
4	N	U	E	S	R

MATRIX-II

	5	6	7	8	9
5	W	O	P	T	I
6	T	I	O	W	P
7	I	W	I	P	T
8	O	P	T	O	W
9	P	T	W	I	O

17. PENT

- (a) 87, 21, 31, 66
(c) 57, 02, 34, 87

- (b) 95, 33, 40, 78
(d) 78, 42, 11, 58

18. NOTE

- (a) 40, 85, 96, 34
(c) 04, 67, 78, 21

- (b) 33, 99, 87, 14
(d) 22, 56, 65, 43

19. GUIDE

MATRIX-I

	0	1	2	3	4
0	1	E	A	O	U
1	A	O	U	I	E
2	E	I	O	U	A
3	O	U	E	A	I
4	U	A	I	E	O

MATRIX-II

	5	6	7	8	9
5	F	D	B	G	H
6	B	G	H	F	D
7	D	F	G	H	B
8	G	H	D	B	F
9	H	B	F	G	D

- (a) 85, 23, 21, 87, 32
(c) 77, 13, 42, 99, 32

- (b) 58, 31, 12, 57, 41
(d) 66, 31, 43, 78, 14

20. CARD

MATRIX-I

	0	1	2	3	4
0	A	B	C	D	E
1	D	C	B	A	E
2	B	A	D	C	E
3	D	B	C	A	E
4	C	D	A	E	B

- (a) 32, 00, 56, 10
(c) 11, 33, 57, 22

MATRIX-II

	5	6	7	8	9
5	P	Q	R	S	T
6	Q	S	P	R	T
7	P	T	R	S	Q
8	Q	S	P	R	T
9	T	P	S	Q	R

- (b) 40, 21, 68, 44
(d) 02, 42, 77, 20

21. CHILD

MATRIX-I

	0	1	2	3	4
0	C	D	E	F	G
1	F	G	C	D	E
2	D	E	F	G	C
3	E	F	G	C	D
4	G	C	D	E	F

- (a) 24, 21, 99, 57, 01
(c) 33, 57, 99, 57, 01

MATRIX-II

	5	6	7	8	9
5	H	K	L	I	N
6	I	N	H	K	L
7	K	L	I	N	H
8	L	I	N	H	K
9	N	H	K	L	I

- (b) 12, 79, 99, 57, 01
(d) 41, 79, 99, 57, 11

22. CALM

MATRIX

0	1	2	3	4	5	6
1	H	R	E	I	P	S
2	S	G	N	D	Z	I
3	B	U	F	T	K	L
4	V	A	P	C	Y	A
5	M	W	C	O	X	N
6	B	A	E	I	L	O

- (a) 44, 62, 65, 51
(c) 53, 54, 51, 31

- (b) 53, 42, 65, 36
(d) 44, 54, 65, 24

23. BEST

MATRIX-I

	0	1	2	3	4
0	B	C	D	E	F
1	E	F	B	C	D
2	C	D	E	F	B
3	F	B	C	D	E
4	D	E	F	B	C

- (a) 24, 21, 77, 97
(c) 24, 22, 77, 96

MATRIX-II

	5	6	7	8	9
5	P	Q	R	S	T
6	S	T	P	Q	R
7	Q	R	S	T	P
8	T	P	Q	R	S
9	R	S	T	P	Q

- (b) 24, 22, 77, 97
(d) 24, 22, 76, 97

24. YEAR

MATRIX-I

	2	3	4	5	6
0	Y	A	H	M	J
1	M	J	H	A	Y
2	A	Y	J	H	M
3	H	J	Y	M	A
4	J	M	A	Y	H

- (a) 23, 27, 15, 61
(c) 34, 31, 32, 28

MATRIX-II

	7	8	9	1	0
2	E	R	V	N	O
3	V	N	O	E	R
4	O	E	R	V	N
5	R	V	N	O	E
6	N	O	E	R	V

- (b) 16, 38, 15, 30
(d) 45, 50, 36, 29

Directions (25-30):

Matrix - I

	0	1	2	3	4
0	I	A	U	E	O
1	E	U	O	A	I
2	O	A	I	E	U
3	E	U	A	O	I
4	E	I	O	A	U

Matrix - II

	5	6	7	8	9
5	K	R	L	M	N
6	M	R	K	N	L
7	K	N	M	L	R
8	M	L	K	R	N
9	N	R	L	K	M

25. MONK

- (a) 58, 33, 67, 98
(c) 65, 04, 89, 75

26. NAIL

- (a) 95, 01, 00, 77
(c) 95, 32, 14, 86

27. LANKA

- (a) 97, 43, 59, 55, 01
(c) 69, 31, 76, 75, 01

28. LIKE

- (1) 57, 14, 76, 10
(3) 78, 33, 87, 03

29. RULE

- (a) 95, 02, 58, 30
(c) 66, 11, 78, 41

30. KRONE

- (a) 87, 79, 03, 89, 23
(c) 87, 56, 04, 76, 23

Direction (31-35):

MATRIX-I

	0	1	2	3	4
0	A	P	O	R	E
1	R	E	P	O	A
2	O	A	E	P	R
3	E	O	R	A	P
4	P	R	A	E	O

MATRIX-II

	5	6	7	8	9
5	F	N	M	S	T
6	T	M	F	N	S
7	M	S	T	F	N
8	N	F	S	T	M
9	S	T	N	M	F

31. MORT

- (a) 66, 21, 32, 77
(c) 57, 13, 03, 88

32. FASE

- (a) 67, 33, 95, 43
(c) 86, 42, 77, 22

33. SERAF

- (a) 69, 04, 10, 21, 88
(c) 76, 43, 24, 32, 99

34. PANT

- (a) 12, 33, 69, 77
(c) 01, 21, 56, 66

35. TORN

- (a) 96, 13, 24, 69
(c) 65, 20, 32, 79

Directions (36-38):

MATRIX-I

	0	1	2	3	4
0	D	O	B	A	I
1	O	B	A	I	D
2	B	A	I	D	O
3	A	I	D	O	B
4	I	D	O	B	A

MATRIX-II

	5	6	7	8	9
5	W	N	R	M	L
6	N	R	M	L	W
7	R	M	L	W	N
8	M	L	W	N	R
9	L	W	N	R	M

36. DRAW

- (a) 41, 66, 23, 55
(c) 23, 57, 30, 68

37. BAND

- (a) 43, 21, 97, 33
(c) 34, 44, 66, 14

38. From the given alternatives select the word which cannot be formed using the letters of the given word : (INTERNATIONAL)

- (a) ALONE
(c) LATER

39. Out of the following options what will be the code of the word 'SUN' ?

K	3	8	9	7
U	6	1	4	8
S	7	6	2	3
N	4	0	8	5

- (a) 7, 4, 0
(c) 0, 6, 4

40. Out of the following options what will be the code for the word 'BAC' ?

A	5	9	8	7
B	8	6	9	7
C	7	4	2	8
D	2	3	8	9

- (a) 2, 5, 7
(c) 9, 8, 3

Distinct Questions

Directions (41 - 43): Some columns are given below and questions are based on it. Certain numbers are given in them.

Which of these lines are related and how?

41.

Line	Column				
	I	II	III	IV	V
1	9	11	18	27	3
2	5	2	10	15	5
3	7	8	14	21	12
4	3	10	6	9	22
5	11	14	22	33	34

- (a) column I, II and III (b) column I, III and V
(c) column II, III and IV (d) column I, III and IV

42.

Line	Column				
	I	II	III	IV	V
1	6	12	24	36	54
2	8	16	24	48	72
3	5	15	25	30	45
4	9	18	36	54	81
5	10	20	40	60	90

- (a) column I, II and V (b) column I, IV and V
(c) column II, III and IV (d) column I, III and V

43.

Line	Column				
	I	II	III	IV	V
1	5	20	25	27	52
2	7	40	49	50	94
3	8	32	64	108	46
4	9	27	81	35	18
5	4	19	16	49	61

- (a) column I, III and V (b) column II, III and IV
(c) column I, II and III (d) column II, IV and V

Directions (44 - 47): Below, there is a table of examinee and various examination centre. In the table, from the left two digit of number are the code of examination centre. Rest four digit are the roll no. of examinee.

432410	526214	433809	613184
453041	643182	562141	648932
521861	613809	463179	563809
561568	431648	433665	526299
641689	453814	613190	623224
433488	433214	648179	523879

44. Which of the examinee having smallest number?
(a) 56 (b) 61
(c) 62 (d) 64
45. In which centre are there maximum number of examinee?
(a) 43 (b) 45
(c) 64 (d) 62
46. In which center-pair are there maximum same number of examinee?
(a) 43-45 (b) 56-61
(c) 52-56 (d) 43-64
47. In which centre-pair are there last two digits are equal of number?
(a) 45-56 (b) 64-61
(c) 43-52 (d) 56-43

Direction (48-50): Given below are some question of sets, among the four sets. Find out the set, which is like the given set.

48. Given Set : (15, 19, 16)
(a) (17, 22, 19) (b) (25, 30, 28)
(c) (16, 20, 17) (d) (32, 35, 31)
49. Given Set : (826, 574, 952)
(a) (764, 386, 854) (b) (842, 654, 928)
(c) (576, 478, 754) (d) (682, 196, 934)
50. Given Set : (48, 84, 128)
(a) (33, 65, 105) (b) (36, 70, 112)
(c) (22, 52, 90) (d) (56, 96, 144)

Previous year Questions (Memory Based)

Directions (1-2):

MATRIX-I

	0	1	2	3	4
4	A	F	K	P	U
3	F	K	A	U	P
2	P	U	F	K	A
1	K	P	U	A	F
0	U	A	P	F	K

MATRIX-II

	5	6	7	8	9
9	D	I	N	S	X
8	X	S	I	D	N
7	N	X	S	I	D
6	S	D	X	N	I
5	I	N	D	X	S

1. SIX-KIDS
(a) 86, 87, 99—40, 41, 86, 64
(b) 98, 96, 85—42, 78, 88, 77
(c) 77, 69, 76—22, 95, 28, 31
(d) 65, 55, 67—05, 25, 91, 40
2. NASA
(a) 97, 32, 34, 45 (b) 56, 13, 98, 40
(c) 03, 14, 69, 98 (d) 97, 35, 43, 58

3. BEE

MATRIX-I

	1	2	3	4	5
1	A	B	C	D	E
2	E	D	A	B	C
3	B	C	D	E	A
4	D	A	E	C	D
5	C	E	B	A	B

- (a) 12, 15, 33
(c) 12, 15, 41

MATRIX-II

	6	7	8	9	10
6	F	G	H	I	J
7	J	I	G	H	F
8	F	H	I	J	G
9	G	J	F	G	I
10	H	E	J	F	E

- (b) 21, 12, 22
(d) 12, 21, 15

Directions (4 - 8):

MATRIX-I

	0	1	2	3	4
0	I	L	A	D	C
1	D	C	I	L	A
2	A	I	D	C	L
3	L	A	C	I	D
4	C	D	L	A	I

MATRIX-II

	5	6	7	8	9
5	W	R	T	E	Y
6	R	W	Y	T	E
7	E	Y	W	R	T
8	Y	T	E	W	R
9	T	E	R	Y	W

4. DAILY

- (a) 10, 31, 21, 42, 85
(c) 41, 14, 13, 24, 98

5. RATE

- (a) 89, 20, 96, 69
(c) 78, 14, 95, 58

6. CIAT

- (a) 11, 21, 31, 87
(c) 23, 34, 43, 95

7. WEAR

- (a) 77, 87, 32, 89
(c) 88, 69, 21, 65

8. DWART

- (a) 10, 77, 32, 89, 68
(c) 34, 99, 43, 66, 79

Directions (9 - 13):

MATRIX-I

	0	1	2	3	4
0	E	H	J	O	A
1	H	O	E	A	J
2	O	J	A	E	H
3	J	A	O	H	E
4	A	E	H	J	O

MATRIX-II

	5	6	7	8	9
5	L	F	S	K	T
6	T	K	L	S	F
7	S	L	F	T	K
8	K	S	T	F	L
9	F	T	K	L	S

9. JOLE

- (a) 21, 44, 89, 34
(c) 14, 33, 98, 23

10. FATH

- (a) 69, 13, 79, 24
(c) 88, 21, 87, 42

11. KHOLE

- (a) 66, 24, 20, 76, 24
(c) 97, 33, 12, 89, 34

12. FEAST

- (a) 77, 00, 14, 86, 96
(c) 69, 23, 31, 75, 87

13. LOHA

- (a) 98, 20, 42, 04
(c) 67, 12, 24, 31

Directions (14 - 18):

MATRIX-I

	0	1	2	3	4
0	A	S	U	E	O
1	U	O	E	S	A
2	S	A	O	U	E
3	E	U	A	O	S
4	O	E	S	A	U

MATRIX-II

	5	6	7	8	9
5	C	R	T	P	N
6	N	P	C	T	R
7	P	C	R	N	T
8	T	N	P	R	C
9	R	T	N	C	P

14. TUNE

- (a) 68, 31, 79, 41
(c) 79, 23, 86, 30

15. CROPS

- (a) 89, 95, 11, 75, 20
(c) 76, 88, 40, 67, 13

16. SENT

- (a) 20, 31, 86, 79
(c) 13, 03, 66, 85

17. PART

- (a) 75, 21, 89, 96
(c) 87, 00, 95, 69

18. SPENT

- (a) 13, 66, 25, 86, 96
(c) 34, 87, 41, 59, 85

19. Identify the set for the word 'BEAD'

MATRIX-I

	0	1	2	3	4
0	I	E	A	O	U
1	A	O	U	I	E
2	E	I	O	U	A
3	O	U	E	A	I
4	U	A	I	E	O

MATRIX-II

	5	6	7	8	9
5	F	D	B	G	H
6	B	G	H	F	D
7	D	F	G	H	B
8	G	H	D	B	F
9	H	B	F	G	D

- (a) 75, 14, 20, 57
(c) 81, 41, 20, 57

- (b) 97, 32, 14, 56
(d) 57, 32, 41, 87

20. Identify the set for the word 'NIFE' ?

MATRIX-I

	0	1	2	3	4
0	E	F	G	H	I
1	H	I	E	F	G
2	F	G	H	I	E
3	I	E	F	G	H
4	G	H	I	E	F

MATRIX-II

	5	6	7	8	9
5	L	M	N	O	P
6	O	P	L	M	N
7	M	N	O	P	L
8	P	L	M	N	O
9	N	O	P	L	M

- (a) 95, 30, 32, 43 (b) 95, 30, 31, 44
(c) 57, 42, 31, 43 (d) 57, 41, 32, 43

21. Identify the set for the word 'AELO' ?

MATRIX-I

	0	1	2	3	4
0	A	E	C	B	D
1	C	D	A	E	B
2	B	E	D	C	A
3	D	A	C	B	E
4	B	E	D	A	C

MATRIX-II

	0	1	2	3	4
0	L	M	O	N	P
1	N	P	L	M	O
2	P	M	O	L	N
3	L	N	P	M	O
4	O	N	L	P	M

- (a) 31, 00, 23, 22 (b) 43, 01, 12, 42
(c) 12, 34, 30, 02 (d) 12, 30, 42, 14

22. Identify the set for the word 'STEP' ?

MATRIX-I

	0	1	2	3	4
0	A	E	M	N	P
1	N	P	A	E	M
2	E	M	N	P	A
3	P	A	E	M	N
4	M	N	P	A	E

MATRIX-II

	5	6	7	8	9
5	I	L	R	S	T
6	R	S	T	I	L
7	T	I	L	R	S
8	L	R	S	T	I
9	S	T	I	L	R

- (a) 66, 88, 31, 03 (b) 58, 67, 01, 11
(c) 65, 87, 01, 32 (d) 85, 75, 01, 40

23. Identify the set for the word 'DIRT' ?

MATRIX-I

	0	1	2	3	4
0	D	E	F	I	N
1	I	N	D	E	F
2	E	F	I	N	D
3	N	D	E	F	I
4	F	I	N	D	E

MATRIX-II

	5	6	7	8	9
5	O	P	R	S	T
6	S	T	O	P	R
7	P	R	S	T	O
8	T	O	P	R	S
9	R	S	T	O	P

- (a) 00, 10, 76, 86 (b) 43, 34, 88, 79
(c) 24, 34, 57, 66 (d) 00, 10, 76, 86

24. Identify the set for the word 'PENS' ?

MATRIX-I

	0	1	2	3	4
0	P	W	I	N	S
1	I	S	P	W	N
2	W	N	I	S	P
3	S	P	W	N	I
4	N	I	S	P	W

MATRIX-II

	5	6	7	8	9
5	A	E	R	O	H
6	O	H	A	E	R
7	E	R	O	H	A
8	H	A	E	R	O
9	R	O	H	A	E

- (a) 12, 67, 21, 30 (b) 43, 56, 13, 23
(c) 43, 56, 21, 42 (d) 31, 57, 21, 42

25. Identify the set for the word 'DRIFT' ?

MATRIX-I

	0	1	2	3	4
0	P	I	D	S	T
1	U	N	R	Q	M
2	D	A	E	F	G
3	B	W	V	L	N
4	C	H	J	K	I

MATRIX-II

	5	6	7	8	9
5	F	P	R	S	D
6	N	L	M	T	I
7	A	S	N	P	R
8	D	G	Z	W	O
9	N	C	K	E	S

- (a) 20, 57, 69, 23, 68 (b) 20, 12, 69, 32, 40
(c) 85, 75, 96, 32, 40 (d) 58, 57, 96, 23, 40

26. Which of these columns are related and how?

Line	Column				
	I	II	III	IV	V
1	6	24	24	89	28
2	5	20	44	78	24
3	7	28	30	77	32
4	9	36	36	49	40
5	10	40	42	47	44

- (a) Column I, II and III (b) Column I, II and V
(c) Column II, III and IV (d) Column I, III and IV

Directions (27-29): Below, a table is given. The number of first two digit shows the number of gas agency and last four digits show consumer.

312864	457894	562354	278262
273572	596971	841021	592684
451234	272572	316824	844294
847801	312525	459876	311864
275684	561997	594949	457801
568585	841874	277582	568234

27. In which gas agency are there consumers number is same ?

- (a) 56 and 27 (b) 45 and 31
(c) 59 and 84 (d) 84 and 45

28. In the above table how many consumers are there of odd number?

- (a) 15 (b) 10
(c) 8 (d) 5

29. In above table, in which gas agency are there maximum number of consumers?

- (a) 45 (b) 84
(c) 27 (d) 31

30. Find out the set, which is like the given set.

Given set : (9, 27, 108)

- (a) (15, 45, 90) (b) (8, 32, 96)
(c) (5, 15, 75) (d) (11, 33, 132)

Practice Set Solutions

Directions (1 - 2):

E $\Rightarrow$ 00, 13, 21, 34, 42 N $\Rightarrow$ 04, 11, 22, 33, 40

O $\Rightarrow$ 56, 67, 75, 88, 99 P $\Rightarrow$ 57, 69, 78, 86, 95

T $\Rightarrow$ 58, 65, 79, 87, 96

1. (d);

Option	P	E	N	T
(1)	87	21	31	66
(2)	95	38	40	78
(3)	57	02	34	87
(4)	78	42	11	58

2. (a);

Option	N	O	T	E
(1)	40	75	96	34
(2)	33	99	87	14
(3)	04	67	78	21
(4)	22	56	65	43

Directions (3 - 7): All the possible representations of the letters.

Matrix-I

F $\Rightarrow$ 00, 12, 24, 31, 43

M $\Rightarrow$ 02, 14, 21, 33, 40

O $\Rightarrow$ 01, 13, 20, 32, 44

R $\Rightarrow$ 04, 11, 23, 30, 42

S $\Rightarrow$ 03, 10, 22, 34, 41

Matrix-II

A $\Rightarrow$ 55, 67, 79, 86, 98

D $\Rightarrow$ 57, 69, 76, 88, 95

I $\Rightarrow$ 58, 65, 77, 89, 96

P $\Rightarrow$ 59, 66, 78, 85, 97

T $\Rightarrow$ 56, 68, 75, 87, 99

3. (d);

Option	R	O	A	D
(1)	04	20	55	78
(2)	23	32	98	99
(3)	42	32	79	58
(4)	11	13	67	69

4. (a);

Option	M	O	S	T
(1)	02	13	34	56
(2)	21	00	03	88
(3)	33	20	11	79
(4)	40	44	22	89

Directions (5 - 7): All the possible representations of letters

F = 00, 12, 24, 31, 43

S = 03, 10, 22, 34, 41

O = 01, 13, 20, 32, 44

R = 04, 11, 23, 30, 42

M = 02, 14, 21, 33, 40

5. (d);

Option	F	O	A	M
(1)	24	01	55	22
(2)	43	32	56	33
(3)	12	13	67	28
(4)	00	01	67	33

6. (a);

Option	S	T	O	P
(1)	10	56	44	97
(2)	41	68	01	77
(3)	22	75	32	86
(4)	38	99	42	59

7. (b);

Option	M	O	S	T
(1)	02	31	34	75
(2)	33	44	22	99
(3)	41	01	42	68
(4)	21	32	38	98

8. (b); F O R K $\rightarrow$ 30 86 13 77

9. (c) 10. (d) 11. (a) 12. (b) 13. (c)

14. (b) 15. (d) 16. (d) 17. (d) 18. (a)

19. (a) 20. (c) 21. (b) 22. (a) 23. (b)

24. (a) 25. (c) 26. (c) 27. (a) 28. (b)

29. (d) 30. (c) 31. (c) 32. (a) 33. (b)

34. (d) 35. (c) 36. (d) 37. (b) 38. (d)

39. (a) 40. (b)

Distinct Solutions

41. (d); Column I, III, IV

I	III	IV
9×1	$9 \times 2 = 18,$	$9 \times 3 = 27$
5×1	$5 \times 2 = 10,$	$5 \times 3 = 15$
7×1	$7 \times 2 = 14,$	$7 \times 3 = 21$
3×1	$3 \times 2 = 6,$	$3 \times 3 = 9$
11×1	$11 \times 2 = 22,$	$11 \times 3 = 33$

42. (b); Column I, IV, V

I	IV	V
6×6	36×1.5	54
8×6	48×1.5	72
5×6	30×1.5	45
9×6	54×1.5	81
10×6	60×1.5	90

43. (a); Column I, III, V

I	III	V
5^2	25 interchanges	52
7^2	49 interchanges	94
8^2	64 interchanges	46
9^2	81 interchanges	18
4^2	16 interchanges	61

44. (a); On centre 56, roll no. 1568 is smallest number.

45. (a); On centre 43, there are 6 examinee.

46. (b); On centre 56 and 61, there are same roll no. examinee.

$$56 \rightarrow 3809 \quad 61 \rightarrow 3809$$

47. (c); On centre 43 and 52, the last two digits of examinee roll no. are same.

$$43 \rightarrow 3214 \quad 52 \rightarrow 6214$$

48. (c); As $15 + 4 \rightarrow 19 - 3 \rightarrow 16$

$$\text{Similarly, } 16 + 4 \rightarrow 20 - 3 \rightarrow 17$$

49. (d); As

$$826 \rightarrow 8 + 2 + 6 \rightarrow 16$$

$$574 \rightarrow 5 + 7 + 4 \rightarrow 16$$

$$952 \rightarrow 9 + 5 + 2 \rightarrow 16$$

Similarly,

$$682 \rightarrow 6 + 8 + 2 \rightarrow 16$$

$$196 \rightarrow 1 + 9 + 6 \rightarrow 16$$

$$934 \rightarrow 9 + 3 + 4 \rightarrow 16$$

50. (d); As (48, 84, 128) all numbers are divided by 4

Similarly, (56, 96, 144) all number are divided by 4.

Previous year Solutions (Memory Based)

1. (b) 2. (b) 3. (d) 4. (a) 5. (c)

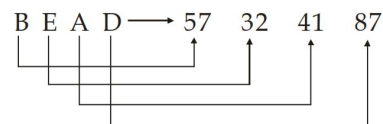
6. (b) 7. (d) 8. (b) 9. (a) 10. (d)

11. (b) 12. (c) 13. (a) 14. (c) 15. (a)

16. (d) 17. (b) 18. (c)

19. (d); The numbers of all letters.

I = 00, 13, 21, 34, 42	E = 01, 14, 20, 32, 43
A = 02, 10, 24, 33, 41	O = 03, 11, 22, 30, 44
U = 04, 12, 23, 31, 40	F = 55, 68, 76, 89, 97
D = 56, 69, 75, 87, 99	B = 57, 65, 79, 88, 96
G = 58, 66, 77, 85, 98	H = 59, 67, 78, 86, 95



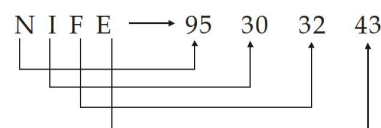
20. (a); The numbers of all letters.

E = 00, 12, 24, 31, 43	F = 01, 13, 20, 32, 44
G = 02, 14, 21, 33, 40	H = 03, 10, 22, 34, 41
I = 04, 11, 23, 30, 42	

L = 55, 67, 79, 86, 98 M = 56, 68, 75, 87, 99

N = 57, 69, 76, 88, 95 O = 58, 65, 77, 89, 96

P = 59, 66, 78, 85, 97



Note :- We can solve these question only by the number of given letter.

21. (c); A = 00, 12, 24, 31, 43 E = 01, 13, 21, 34, 41

L = 00, 12, 23, 30, 42 O = 02, 14, 22, 34, 40

AELO = 12, 34, 30, 02

22. (b); S = 58, 66, 79, 87, 95 T = 59, 67, 75, 88, 96

E = 01, 13, 20, 32, 44 P = 04, 11, 23, 30, 42

STEP = 58, 67, 01, 11

23. (c); D = 00, 12, 24, 31, 43 I = 03, 10, 22, 34, 41

R = 57, 69, 76, 88, 95 T = 59, 66, 78, 85, 97

DIRT = 24, 34, 57, 66

24. (c); P = 00, 12, 24, 31, 43 E = 56, 68, 75, 87, 99

N = 03, 14, 21, 33, 40 S = 04, 11, 23, 30, 42

PENS = 43, 56, 21, 42

25. (a); D = 02, 20, 79, 85 I = 01, 44, 69
 R = 12, 57, 79 F = 23, 55
 T = 04, 68
 DRIFT = 20, 57, 69, 23, 68

26. (b); Column I, II, V
 I II V
 6×4 $24 + 4$ 28
 5×4 $20 + 4$ 24
 7×4 $28 + 4$ 32
 9×4 $36 + 4$ 40
 10×4 $40 + 4$ 44

27. (d); On gas agency 84 and 45, the number is same.
 $84 \rightarrow 7801$ $45 \rightarrow 7801$

28. (c); Odd number consumer are
 7801, 8585, 6971, 2525, 1997, 1021, 4949, 7801

29. (c); Maximum number of consumer are on agency 27.
 3572, 5684, 2572, 7882, 8262

30. (d); As, $9 \times 3 \rightarrow 27 \times 4 \rightarrow 108$
 Similarly, $11 \times 3 \rightarrow 33 \times 4 \rightarrow 132$